

SHILKA, Russian Federation

WMO: 308620

Lat: 51.867N

Lon: 116.033E

Elev: 1959

StdP: 13.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-28.4	-36.9	0.7	-31.5	-33.1	0.9	-27.6	20.5	3.0	18.3	1.0	3.4	270	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	87.9	64.6	84.0	63.4	80.6	62.4	68.0	80.4	66.4	77.6	64.8	75.6	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	96.8	71.9	62.7	91.8	71.0	61.0	86.3	69.5	33.7	80.9	32.3	77.7	31.1	75.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	20.6	17.7	-38.6	94.1	4.8	3.9	-42.0	96.9	-44.9	99.2	-47.5	101.5	-51.0	104.3		
(5)			-38.0	70.7	5.3	2.6	-41.9	72.6	-45.0	74.1	-47.9	75.5	-51.8	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	-10.8	-1.6	16.4	37.0	51.3	64.5	68.6	63.7	51.0	33.5	9.4	-7.8		
	DBStd	29.59	9.99	10.44	12.41	8.89	7.82	6.91	5.20	6.07	7.76	9.27	12.68	11.30		
	HDD50	8453	1885	1444	1040	400	77	1	0	1	80	515	1218	1792		
	HDD65	12518	2350	1864	1505	839	430	90	23	95	421	976	1668	2257		
	CDD50	1683	0	0	0	11	118	436	576	427	111	4	0	0		
	CDD65	270	0	0	0	0	5	74	133	56	2	0	0	0		
	CDH74	2908	0	0	0	6	108	905	1267	566	56	0	0	0		
	CDH80	905	0	0	0	1	22	302	412	161	7	0	0	0		
(14) Wind	WSAvg	7.5	6.0	6.6	7.8	10.2	10.1	7.9	6.9	6.7	7.5	7.7	6.6	5.5		
(15) Precipitation	PrecAvg	12.6	0.1	0.2	0.2	0.5	0.9	2.4	2.8	3.1	1.7	0.4	0.3	0.2		
	PrecMax	18.8	0.4	0.4	0.7	2.1	2.4	6.0	6.2	6.1	4.4	0.8	0.8	0.8		
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.0	0.4	0.5	0.7	0.2	0.0	0.0	0.0		
	PrecStd	2.8	0.1	0.1	0.2	0.5	0.6	1.5	1.5	1.4	1.1	0.3	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	29.7	53.1	71.7	82.9	93.6	93.9	89.3	79.9	65.1	42.1	24.8		
		MCWB	14.9	24.5	39.1	49.6	57.2	66.2	67.6	64.9	58.5	47.6	34.2	21.1		
	2%	DB	11.5	23.1	45.3	63.8	76.5	87.0	88.6	84.5	74.0	57.6	36.5	17.8		
		MCWB	9.4	19.4	34.5	45.0	54.4	63.0	65.6	64.1	56.0	43.7	30.4	15.5		
	5%	DB	7.3	17.9	39.3	57.8	71.9	82.9	84.3	80.3	69.3	52.4	31.9	12.4		
		MCWB	5.6	15.2	30.8	41.7	52.4	61.6	64.8	62.9	54.2	40.8	27.5	10.6		
	10%	DB	3.1	13.6	34.4	52.6	67.0	79.1	80.7	76.1	64.9	47.5	27.1	7.6		
		MCWB	1.7	11.5	27.8	38.9	49.8	60.4	63.9	61.7	52.3	37.8	23.5	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	24.6	40.0	50.3	59.6	68.4	71.1	69.1	61.4	48.1	34.9	21.2		
		MCDB	17.7	29.4	51.9	70.3	78.3	85.7	84.8	82.7	72.7	64.1	41.4	24.8		
	2%	WB	9.5	19.3	34.7	45.5	56.1	65.8	68.9	66.6	58.9	44.5	30.9	15.5		
		MCDB	11.5	23.0	44.4	61.8	72.1	80.1	81.5	77.1	68.0	56.4	35.9	17.8		
	5%	WB	5.7	15.1	31.1	42.3	53.6	64.0	67.3	65.0	56.5	41.3	27.3	10.6		
		MCDB	7.4	17.7	39.1	56.7	69.4	77.5	78.8	75.3	65.8	51.2	31.9	12.4		
	10%	WB	1.8	11.5	27.8	39.2	50.9	62.3	65.8	63.4	53.9	38.5	23.5	6.3		
		MCDB	3.0	13.5	34.0	51.9	65.5	75.0	76.3	73.4	63.6	46.7	27.0	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	18.3	20.4	21.0	23.3	22.6	20.5	20.3	21.3	18.5	15.8	13.5		
		MCDBR	16.2	19.7	22.6	29.4	30.9	29.3	27.4	28.0	28.2	24.8	17.9	15.1		
		MCWBR	14.6	16.9	14.9	16.7	14.7	11.2	10.3	11.1	13.1	14.0	13.3	13.6		
		MCDBR	16.1	19.4	22.5	27.8	27.4	24.4	21.2	21.8	23.4	24.1	17.7	15.1		
	5% WB	MCWBR	14.7	16.7	15.4	16.3	14.3	10.3	9.7	9.8	11.6	14.0	13.4	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.276	0.317	0.416	0.493	0.467	0.419	0.388	0.344	0.306	0.275	0.254		
	taud		2.224	2.186	2.165	2.033	1.875	2.067	2.275	2.343	2.401	2.413	2.289	2.180		
	Ebn at Noon		243	267	276	258	242	251	263	266	266	256	229	215		
	Edh at Noon		21	30	37	48	59	49	40	35	30	23	20	18		
(44) All-Sky Solar Radiation	RadAvg		464	836	1344	1660	1842	1945	1780	1548	1220	826	477	352		
	RadStd		11	17	56	98	122	163	132	96	95	52	21	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page